

Ankur Bambhrolia

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PROFESSIONAL SUMMARY

Data Scientist with an M.S. in Biological Data Science and a B.S. in Biology, using Python, R, SQL, statistical modeling, machine learning, simulation, and visualization to solve complex scientific and operational problems. My background spans biological and healthcare data, manufacturing analytics, quality systems, and large-scale inventory operations, with a particular interest in applying data science to the life sciences. Skilled at turning messy, multi-source data into structured analyses, models, and practical recommendations.

SKILLS

- **Programming & Data:** Python, R, SQL, MySQL, BigQuery, Excel / Power Query
- **Modeling & Analysis:** Statistical modeling, machine learning, Monte Carlo simulation, time-series analysis, PCA, hypothesis testing, forecasting
- **Scientific Computing:** Genomics, biological modeling, Unix / shell, Jupyter
- **Python:** pandas, NumPy, Matplotlib, scikit-learn, scikit-bio, PyTorch
- **R:** tidyverse, dplyr, ggplot2
- **Visualization & Reporting:** Power BI, Tableau, R Markdown, Quarto
- **Development & Data Formats:** Git / GitHub, XML

DATA SCIENCE PROJECTS

Surgical Ramp-Up Capacity Simulation | *Python, SQL, Monte Carlo Simulation*

July 2026

Healthcare Operations Project | *Python, SQL, Time-Series Analysis*

Dallas-Fort Worth, TX

- Analyzed elective surgical volume, inpatient admissions, discharge patterns, and length-of-stay data to evaluate how incremental procedures could affect hospital capacity.
- Developed Monte Carlo stress-test scenarios for phased surgical growth under normal, elevated LOS, elevated ED admission, and combined-risk conditions.
- Used historical inpatient census thresholds of 33 and 42 patients to quantify capacity risk and distinguish routine variation from high-pressure scenarios.
- Translated simulation results into phased scheduling and capacity-planning recommendations while documenting assumptions and operational limitations.

scikit-bio Open Source Contributions | *Python, NumPy, PyTorch, Array API*

March 2026

Open-Source Scientific Computing / Bioinformatics

Dallas-Fort Worth, TX

- Contributed to **scikit-bio**, an open-source scientific Python library for bioinformatics, with work focused on improving compatibility with the Python **Array API** standard.
- Refactored scientific-computing functions to better support multiple array backends, including **NumPy** and **PyTorch**, reducing assumptions tied to a single numerical-computing library.
- Developed and updated tests to validate behavior across supported array types, helping improve reliability and consistency of scientific workflows across different computational backends.
- Worked within an established open-source codebase using Git/GitHub, code review, testing, and existing project conventions, building experience contributing to mature scientific software.

Vibrio vulnificus Risk Analysis & Forecasting

January 2025 — May 2025

Capstone Project | *Python, SQL, Time-Series Analysis*

Tempe, AZ

- Integrated 20+ years of CDC surveillance data with NOAA sea-surface temperature and NASA salinity data to build a harmonized longitudinal dataset for microbial and environmental risk analysis.
- Cleaned, validated, and temporally aligned heterogeneous biological and environmental datasets while evaluating missingness and data-quality limitations.
- Conducted exploratory, correlational, and time-series analyses and evaluated multiple forecasting approaches using holdout validation and error metrics.
- Developed interactive visualizations communicating historical patterns, seasonality, environmental associations, projected risk, and key sources of uncertainty.

PROFESSIONAL EXPERIENCE

3PL Coordinator / Operations Data & Inventory Reconciliation

April 2026 — Present

Ventura Foods LLC

Saginaw, TX

- Reconcile more than \$26M in inventory across JDE, AS400, EDI, and third-party warehouse systems, using large CSV extracts, composite-key matching, pivot tables, and exception analysis to identify missing, duplicated, and inconsistent records.
- Reduced net inventory discrepancy from approximately \$20K to \$5K within three months by tracing inconsistencies across order, receipt, shipment, hold, and inventory records and coordinating corrective actions with warehouse, Finance, Inventory, and technical stakeholders.
- Built and maintained a structured exception-tracking workflow to classify transaction failures, quantify financial impact, document resolution status, and prioritize discrepancies requiring investigation.
- Investigate 944 and 945 EDI transaction failures by comparing expected and observed system activity, reducing recurring exceptions and improving transaction closeout reliability.

Senior Manufacturing Coordinator (Data Analytics)

November 2024 — April 2026

Ventura Foods LLC

Saginaw, TX

- Applied exploratory analysis, trend monitoring, principal component analysis, and data visualization to support root-cause investigations, contributing to approximately 20% fewer repeat quality events and 25% faster investigations.
- Cleaned, standardized, and validated data across multiple production lines while developing recurring reports and dashboards used by Quality, R&D, and Operations to monitor performance and evaluate interventions.
- Partnered with subject-matter experts across Quality Assurance, R&D, Operations Excellence, and production to translate loosely defined operational questions into structured analyses and recommended actions.
- Led production test runs and supported continuous-improvement and customer programs for McDonald's and Burger King, coordinating Quality, R&D, Operations, production, and customer-facing communication from testing through follow-up actions.

Quality Assurance Coordinator/Quality Assurance Tech II

April 2021 — November 2024

Ventura Foods LLC

Saginaw, TX

- Validated and maintained high-volume microbiological and quality datasets covering approximately 300 samples per day, ensuring records were complete, traceable, and suitable for monitoring and investigation.
- Investigated missing, inconsistent, and irregular records across laboratory testing, product holds, deviations, and documentation workflows, coordinating corrections with Quality, laboratory, and Operations teams.

EDUCATION

Master of Science (Biological Data Science)

May 2025

Arizona State University (GPA: 4.0)

Tempe, AZ

- **Selected Courses:** Functional Genomics, Biological Statistics, Applied Mathematics in Biology, biological modeling in Python, R programming, Unix/shell computing

Bachelor of Science (Biology)

May 2020

Texas Tech University (GPA: 3.6)

Lubbock, TX